NRC Research Press

ARTICLE

Habitat loss, fragmentation, and the genetic status of Roanoke bass

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Abstract: Roanoke bass (*Ambloplites cavifrons*) persist in five river basins in the eastern US, where they are threatened by invasive species, habitat loss and degradation, and hydrologic fragmentation. We conducted the first conservation genetic study of *A. cavifrons*, analyzing variation at 19 nuclear microsatellite DNA loci and the cytochrome *b* mitochondrial DNA gene to estimate population structure and demography, genetic relationships among populations, and the role of landscape features in regulating genetic diversity and differentiation. Most streams harbored genetically distinguishable populations, with high connectivity among reaches within streams but no contemporary dispersal among streams. In contrast, mitochondrial divergence within and among basins was weak, suggesting historically higher range-wide connectivity. Most populations exhibited small effective population sizes and evidence of past population bottlenecks. Genetic diversity correlated positively with patch size but negatively with watershed urban and agricultural development, suggesting that habitat loss, degradation, and fragmentation have acted in concert to reduce population viability. Mitigating these impacts may require a combination of tactics, including restoring habitat, limiting the spread of invasive competitors, and reestablishing historical connectivity.

Résumé : Le crapet (*Ambloplites cavifrons*) persiste dans cinq bassins hydrographiques de l'est des États-Unis, où il est menacé par des espèces envahissantes, la perte et la dégradation d'habitats et la fragmentation hydrologique. Nous avons mené la première étude de génétique de la conservation d'*A. cavifrons*, en analysant les variations en 19 sites microsatellites d'ADN nucléaire et le gène du cytochrome *b* de l'ADN mitochondrial afin d'estimer la structure et la démographie des populations, les relations génétiques entre ces populations et le rôle d'éléments du paysage dans la régulation de la diversité et de la différenciation génétiques. La plupart des cours d'eau présentent des populations pouvant être distinguées génétiquement, une forte connectivité caractérisant les tronçons d'un même cours d'eau, mais aucune dispersion contemporaine n'étant relevée d'un cours d'eau à l'autre. En revanche, la divergence mitochondriale au sein de bassins et entre ceux-ci est faible, ce qui indiquerait une plus grande connectivité passée à l'échelle de l'aire de répartition. La plupart des populations présentent de petites tailles effectives et des indices de goulots d'étranglement génétiques passés. La diversité génétique est positivement corrélée à la grandeur de la parcelle, mais négativement corrélée à l'aménagement urbain et agricole dans le bassin versant, ce qui indiquerait que la perte, la dégradation et la fragmentation des habitats ont agi de concert pour réduire la viabilité des populations. L'atténuation de ces impacts pourrait nécessiter une combinaison d'approches, dont la restauration d'habitats, la limitation de la propagation de concurrents envahissants et le rétablissement de la connectivité historique. [Traduit par la Rédaction]

Introduction

Recent reports indicate that freshwater taxa have declined by 83% since 1970, making freshwater habitats the most threatened ecosystems on earth (WWF 2018). The main drivers of this decline are the loss, degradation, and fragmentation of freshwater habitat, which act in concert to elevate extinction risk (Dudgeon et al. 2006; Jelks et al. 2008). Freshwater habitats often are naturally isolated, which has contributed to the high endemism of freshwater taxa (Allan and Flecker 1993), but this fragmentation has been exacerbated by human activities such as dam construction. In addition to being physical barriers to fish movement, dams can act as functional barriers by reducing the quality of stream habitat (Roberts et al. 2013). Heightened isolation of fish populations may increase genetic drift and decrease genetic diversity (DeHaan et al. 2012), increase the rate of inbreeding (Mills and Allendorf 1996), and prevent recolonization following extirpation (Meldgaard et al. 2003). Isolated fish populations are further threatened by reduced habitat quality in remnant habitat patches. Agricultural and urban development often lead to increases in sedimentation, nutrient enrichment, and chemical pollution and may cause increases in stream temperature and loss of large woody debris (Allan 2004). Reductions in habitat area and quality may drive a feedback loop with dam-driven fragmentation, further isolating fish populations into fewer, smaller habitat patches.

Endemic to the Roanoke, Dan, Chowan, Tar, and Neuse basins of Virginia and North Carolina, the Roanoke bass (*Ambloplites cavifrons*) has one of the smallest native ranges of any sportfish in the eastern US, and this range appears to be shrinking (Jenkins and Burkhead 1994; Eschenroeder and Roberts 2018). The species occurs in a region long affected by row-crop agriculture and river impoundment, and increasingly affected by urbanization (Jenkins and Burkhead 1994), but it is unknown what effect this has had on population structure, demography, and viability. Past distributional surveys have indicated that *A. cavifrons* populations were disjunct as a result of patchiness of suitable habitat (Petrimoulx 1983), and this patchiness has likely been exacerbated as a result of changes to streams over the past century. The range of *A. cavifrons*

Received 21 March 2019. Accepted 10 July 2019.

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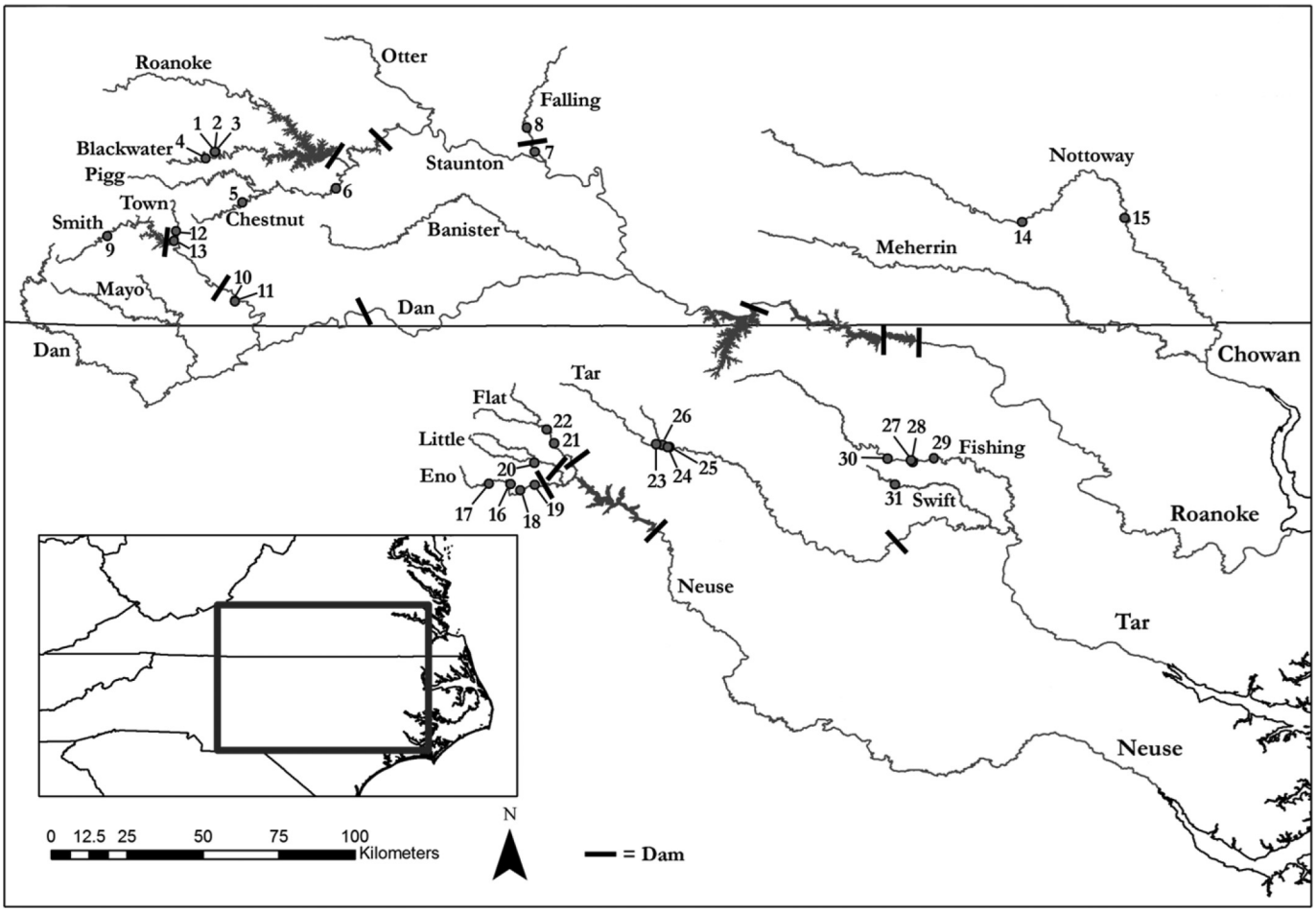
Published at www.nrcresearchpress.com/cjfas on 19 July 2019.

Fig. 1. Map of the study area. Numbered circles indicate sample sites and correspond with the codes in Table 1. Several adjacent sites (e.g., 1, 2, and 3) appear as a single point due to the scale. Solid black bars indicate major dams between sites. Map created in ArcMap version 10.4 (ESRI 2016).



has been further reduced through the introduction of rock bass (*Ambloplites rupestris*), which has led to the replacement of *A. cavifrons* by *A. rupestris* in several streams (Eschenroeder and Roberts 2018; Jenkins and Cashner 1983; Cashner and Jenkins 1982). The largest remaining sport fisheries for *A. cavifrons* are located in the Nottoway and Meherrin rivers in Virginia and the Eno River in North Carolina (Eric Brittle, Virginia Department of Game and Inland Fisheries (VDGIF), personal communication; Corey Oakley, North Carolina Wildlife Resource Commission (NCWRC), personal communication).

Effective management of *A. cavifrons* requires an understanding of the structure, size, and genetic diversity of populations, as well as historical and contemporary genetic and demographic connections between populations. Analysis of variation at genetic markers can provide insight into all of these phenomena (Manel et al. 2003; Storfer et al. 2007; Luikart et al. 2010) and shed light on how natural and anthropogenic landscape features interact to sculpt the demography and evolution of populations (Castric et al. 2001; Yamamoto et al. 2004). Understanding such influences allows for the delineation of management units and design of translocation programs to mimic historical connectivity (Frankham et al. 2011; Vrijenhoek 1998; Mills and Allendorf 1996).

In this study, we applied a suite of complementary analyses of nuclear and mitochondrial DNA marker data collected from populations throughout the range of *A. cavifrons* to address four main questions critical to conserving the species: (1) What is the contemporary population genetic structure and what maintains population boundaries? (2) What is the extent of contemporary and historical gene flow among populations? (3) How do genetic diversity and effective population size vary among populations, and to what extent have modifications of the riverscape affected genetic diversity? (4) Based on this information, how might *A. cavifrons* best be conserved?

Methods

Sample collection

Personnel from Virginia Tech, VDGIF, and NCWRC collected 568 *Ambloplites* specimens from 45 locations across 17 Virginia and North Carolina watersheds between 2012 and 2016 (Fig. 1; Table 1). Fish were collected by angling or by backpack, barge, or boat electrofisher, depending on site conditions. Sites were distributed across all North Carolina and Virginia watersheds known to historically harbor *A. cavifrons*, with the exception of the Meherrin River in Virginia, which was not sampled due to low-flow conditions that precluded access by a boat electrofisher. Upon capture, a fin clip was removed and placed in either a coin envelope or a vial of 95% ethanol. Tissue samples were stored at −20 °C until DNA extraction. Based on results from Eschenroeder and Roberts (2018), 331 of these individuals, originating from 31 sites across 12 Virginia and North Carolina watersheds, were determined to be genetically pure *A. cavifrons* (others were *A. rupestris* or hybrids) and were further analyzed for this study.

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Table 1. Locations of 31 sites where *A. cavifrons* were sampled for this study.

Basin	Watershed	Stream segment	Latitude	Longitude	*n*	Map code	Site code
Roanoke	Blackwater	Blackwater River	37.0528	−79.8824	6	1	BW1
		Blackwater River	37.0541	−79.8819	9	2	BW3
		Blackwater River	37.0541	−79.8823	7	3	BW2
		Blackwater River	37.0345	−79.9094	1	4	BW4
	Pigg	Chestnut Creek	36.9063	−79.8010	3	5	CHEST
		Lower Pigg River	36.9468	−79.5249	1	6	PIGG
	Falling	Lower Falling River	37.0540	−78.9354	5	7	LFALL
		Upper Falling River	37.1268	−78.9595	21	8	UFALL
	Smith	Upper Smith River	36.8053	−80.2008	10	9	USMITH
		Lower Smith River	36.6138	−79.8226	2	10	LSMITH1
		Lower Smith River	36.6141	−79.8225	2	11	LSMITH2
		Town Creek	36.8210	−79.9966	10	12	TOWN1
		Town Creek	36.7925	−80.0027	12	13	TOWN2
Chowan	Nottoway	Nottoway River	36.8477	−77.4934	18	14	NOTT1
		Nottoway River	36.8590	−77.1898	28	15	NOTT2
Neuse	Eno	Eno River	36.0751	−79.0076	15	16	ENO1
		Eno River	36.0757	−79.0708	9	17	ENO2
		Eno River	36.0568	−78.9784	18	18	ENO3
		Eno River	36.0723	−78.9357	3	19	ENO4
	Little	Little River	36.1378	−78.9371	33	20	LITTLE
	Flat	Flat River	36.1957	−78.8781	37	21	FLAT1
		Flat River	36.2357	−78.9001	12	22	FLAT2
Tar	Tar	Tar River	36.1907	−78.5577	24	23	TAR1
		Tar River	36.1840	−78.5369	1	24	TAR2
		Tar River	36.1932	−78.5758	2	25	TAR3
		Tar River	36.1839	−78.5419	3	26	TAR4
	Fishing	Fishing Creek	36.1448	−77.8215	9	27	FISH1
		Fishing Creek	36.1411	−77.8174	21	28	FISH2
		Fishing Creek	36.1515	−77.7537	2	29	FISH3
		Fishing Creek	36.1498	−77.8917	2	30	FISH4
	Swift	Swift Creek	36.0741	−77.8691	5	31	SWIFT
Total					331		

Note: The number (*n*) of individuals sampled and genotyped per site varied based on fish density and capture efficiency.

Laboratory methods

DNA was extracted from fin clips using a DNeasy Blood and Tissue Kit (Qiagen, Hilden, Germany) following the manufacturer's protocols. Using libraries developed from *A. rupestris*, we previously identified seven microsatellite loci that were polymorphic in *A. cavifrons* (Eschenroeder and Roberts 2016). To facilitate more powerful genetic analyses of *A. cavifrons*, we partnered with the Savannah River Ecology Laboratory (SREL) to develop an additional panel of microsatellite markers for that species. Using paired-end Illumina sequencing, SREL identified an additional 7453 potential microsatellite loci; from these, we identified 12 that reliably amplified and exhibited polymorphism in *A. cavifrons* samples (refer to online Supplementary File 1[1]). The 19 total markers were grouped into five multiplexes for PCR (Supplementary File 1[1]). Amplified PCR products were separated in an ABI 3500 Genetic Analyzer and sized in GeneMapper version 4.0 (Applied Biosystems, Foster City, California, USA).

In addition to the 19 nuclear markers, we analyzed variation at the mitochondrial cytochrome *b* gene across a sample of 2 to 10 individuals (five, in most cases) from each deduced population or discrete stream segment (64 sequences total) using the degenerate primers AmbloCytB-F (5′-RTGRCTTGAAAAACCACCGTTG) and AmbloCytB-R (5′-CYCSRYVTCCRGTTTACAAGAC). PCR reaction mixes were 25 μL in total volume and consisted of 12.5 μL GoTaq Mastermix (Promega, Madison, Wisconsin, USA), 0.5 μL forward primer, 0.5 μL reverse primer, 9.5 μL molecular-grade H_20, and 2 μL of template DNA (typical concentration ~40 ng·μL^{-1}). Cycling conditions were as follows: initial denaturation at 95 °C (150 s), 35 cycles of denaturation at 95 °C (60 s), annealing at 58 °C (60 s), and extension at 72 °C (90 s), followed by a final extension at 72 °C (420 s). PCR products were cleaned with ExoSAP-IT (Affymetrix, Santa Clara, California, USA) and forward- and reverse-sequenced by Eton Bioscience, Inc. (Research Triangle Park, North Carolina, USA). Forward and reverse fragments were checked and aligned in Sequencher version 5.4. software (Gene Codes Corporation, Ann Arbor, Michigan, USA), and consensus sequences were aligned using the "Clustal" option in MEGA version 6 (Tamura et al. 2013). All sequences were trimmed to a 973 base-pair fragment that reliably sequenced in all 64 individuals.

Nuclear data analysis

Prior to preliminary tests of microsatellite marker performance, individuals were grouped into 13 tentative population groups based on the distribution of putative movement barriers (dams and watershed boundaries) between sites (Supplementary File 2[1]). Using ARLEQUIN version 3.5 (Excoffier and Lischer 2010), we tested whether genotype frequencies were at Hardy–Weinberg equilibrium for each locus in each population (10^6 Markov chain Monte Carlo (MCMC) steps following a burn-in of 10^5 steps) and tested for linkage disequilibrium between each pair of loci within each population (10^5 permutations). A sequential Bonferroni correction was used to adjust Hardy–Weinberg and linkage disequilibrium test statistics for multiple comparisons to achieve an experiment-wide alpha of 0.05 for each family of tests (Rice 1989).

Following these preliminary tests, we used further analyses to test hypotheses about the spatial scale and drivers of population

[1]Supplementary data are available with the article through the journal Web site at http://nrcresearchpress.com/doi/suppl/10.1139/cjfas-2019-0103.

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structure and grouped individuals into data-informed, post hoc populations. We utilized the Bayesian clustering approach described by Pritchard et al. (2000) implemented in the software STRUCTURE 2.3.4 to estimate the number of populations (K) that produced the sample of genotypes. The model allowed for admixture and correlation of allele frequencies among clusters and searched parameter space using 10^6 recorded MCMC steps following a burn-in of 10^5. A preliminary analysis of K values from 1 to 31 (data not shown) indicated that there was no improvement in the log-likelihood of the model beyond a K of 20. We therefore ran ten model replicates for each K value from 1 to 20. To remove the effect of occasional outlier replicates that failed to converge and had unusually low log-likelihood values, for each K value we discarded the replicate with the lowest log-likelihood and then calculated mean log-likelihood across the remaining nine replicates, using this statistic as a measure of support for alternative K values. For comparison, we also evaluated support for alternative K values based on Evanno et al.'s (2005) delta-K statistic, calculated using the formulas in that paper. We followed analysis of the full dataset by analyzing each of the K subsets individually to determine whether hierarchical population structure existed that went undetected in the first analysis (Janes et al. 2017). For each subset analysis, we evaluated support for K_{subset} values of 1 to 5; all other model settings were as described above.

In addition to individual-based STRUCTURE analyses, we used three group-based approaches to further investigate population structure and its causes. These analyses were conducted at the site scale, using sites from which at least five individuals were sampled. To visualize population structure and genetic relationships among sites, we calculated Nei's minimum genetic distance (D_m; Nei 1973) among pairs of sites and used D_m values to create a neighbor-joining tree in POPULATIONS 1.2.3 (Langella 1999), then visualized the tree in Figtree (Rambaut and Drummond 2009). To estimate the primary spatial scales at which populations were structured, we conducted an analysis of molecular variance (AMOVA; Excoffier et al. 1992) in ARLEQUIN. Molecular variation was decomposed into four hierarchical spatial scales: among basins, among streams within basins, among sites within streams, and among individuals within sites. Permutation tests (10^6 iterations) assessed whether the percentage of variation at each scale was significantly different from zero.

In the third approach, we used the Bayesian generalized linear mixed modeling (BGLMM) framework outlined by Roberts et al. (2013) to assess the relative roles of distance-attenuated dispersal (i.e., isolation by distance (IBD)), physical movement barriers, and genetic drift in promoting spatial population structure among sites. These BGLMMs contained random effects that accounted for nonindependence of multiple comparisons involving the same site. The response variable was F_{ST}, estimated among all pairs of sites (190 comparisons) in ARLEQUIN. Candidate predictors of pairwise F_{ST} included (*i*) the spatial distance between sites (measured along stream channels in ArcMap version 10.4; ESRI 2016), as expected under IBD, (*ii*) the presence or absence of a major dam or basin boundary between pairs of sites, and (*iii*) the mean genetic diversity (H_E) of pairs of sites, centered across all site comparisons to have a mean of zero. The latter factor enabled us to parse the effects of genetic drift (e.g., due to small or variable population size) from the effects of migration. We hypothesized that spatial distance and barrier presence would be positively related to F_{ST}, but that diversity would be negatively related to F_{ST}.

BGLMM models were fit using the MCMCglmm package (Hadfield 2010) in R 2.13.2 (R Core Team 2012). Models assumed that F_{ST} was Gaussian-distributed and included two linked random effects that accounted for the two sites being compared in a given observation. We used diffuse priors for all parameters (mean = 0, variance = 10^{10}) and estimated posteriors based on 1000 MCMC iterations sampled from a total of 20 000 iterations (burn-in = 10 000; thinning interval = 10). Examination of MCMC trace plots indicated that models achieved stationarity, and replicate model runs produced nearly identical posterior estimates (not shown). A total of 13 alternative models were evaluated, consisting of all possible combinations of the fixed effects (spatial distance, barrier presence, and genetic diversity) and their interactions (distance × barrier, diversity × barrier), plus an intercept-only null model. We assessed relative support for competing models based on model weights calculated from the deviance information criterion (DIC; Spiegelhalter et al. 2002), with smaller DIC values and greater weights indicating greater support. We considered any model with DIC weight > 0.01 worthy of interpretation and included it in our confidence set. For prediction, parameter estimates were averaged across all models in the confidence set, weighted by model weights. Based on composite parameter estimates, model goodness-of-fit was assessed based on pseudo-R^2, the squared correlation between the observed- and model-predicted values of F_{ST} (Efron 1978).

After the delineation of populations by the above analyses, we assessed the prevalence of apparent dispersal between populations. We used STRUCTURE to estimate the posterior probability that each individual's genotype originated from the population where it was collected, versus any of the other sampled populations. An individual whose resident probability was low relative to its immigrant probability was presumed to be a first-generation migrant from the population with the highest probability. For this analysis, K was fixed at the estimated true number of populations, as determined by the previously described STRUCTURE analysis of the full dataset, and capture location was incorporated as a Bayesian prior in the model. The model assumed correlation of allele frequencies among populations and a liberal prior migrant probability of 0.5, which gave every individual an equal prior probability of being a resident or immigrant. Use of alternative, more conservative migrant probabilities (<0.5) produced identical model outcomes (see Results). Three replicate runs were performed, each searching parameter space for 10^6 recorded MCMC steps, following a burn-in of 10^5. All individuals were included in this analysis, except for those captured at sites in Chestnut Creek, lower Pigg River, and lower Smith River (n = 8 individuals total), as these sites were represented by too few individuals to characterize those populations' allele frequencies.

For each of the post hoc populations delineated by the STRUCTURE analysis, we estimated a suite of genetic diversity statistics. Unbiased gene diversity (H_E) and allelic richness (A_4; rarified to the minimum sample size of four individuals) were estimated in FSTAT 2.9.3.2 (Goudet 2001), whereas the ratio of the number of alleles to the size range of alleles (M) was estimated in ARLEQUIN. Garza and Williamson (2001) have shown that low values of M are indicative of past bottlenecks. Estimated M values were compared with the published ranges of M values from populations known to have experienced declines and populations known to have been stable (Garza and Williamson 2001). To assess the size and potential viability of populations, we estimated effective population size (N_e) using the linkage disequilibrium method implemented in LDN_e version 1.31. In the LDNe approach, rare alleles can have a large effect on estimates (Waples and Do 2008). We estimated mean N_e under two different thresholds for excluding rare alleles: 0.05 and 0.02. Ninety-five percent confidence intervals of N_e were estimated by jackknifing. M and N_e were calculated only for populations from which at least 20 individuals were sampled.

We next tested the hypotheses that population genetic diversity would be positively related to the habitat area and habitat quality of areas where populations occurred. For all populations with a sample size of at least 20 individuals, we correlated each of three diversity statistics (H_E, A_4, and M) with each of seven landscape indices. To index habitat size, we estimated watershed area upstream of the downstream-most site sampled in each population, using StreamStats version 3 (US Geological Survey 2012; water.usgs.gov/osw/streamstats/). As indices of potential habitat quality,

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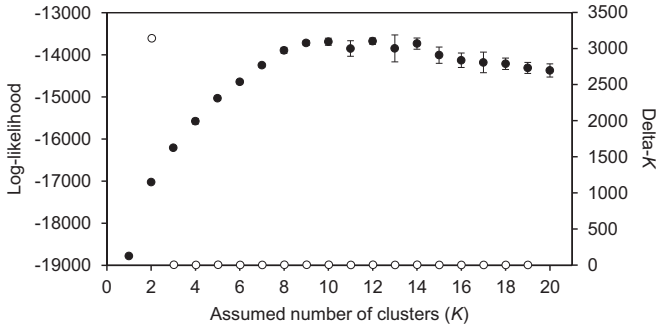
Fig. 2. Relationship between the assumed number of genetic clusters (*K*) and the mean log-likelihood (closed circles) and delta-*K* statistic (open circles), based on nine replicate models run in STRUCTURE for each value of *K* between 1 and 20. Error bars show 95% confidence intervals across replicate model runs.



we used StreamStats to calculate three measures of contemporary watershed land use and three measures of the rate at which land use has changed in the recent past. Contemporary land use was based on the 2011 National Landcover Database (NLCD), whereas rate-of-change was calculated as the percent change in each land-cover type between the 2001 and 2011 NLCDs (Homer et al. 2007, 2015). Three land-cover types were recognized: percent forested area (NLCD classes 41, 42, and 43), percent developed area (classes 21, 22, 23, and 24), and percent cropland area (class 82). For populations sampled at multiple sites, we averaged land cover percentages across sites for that population. We hypothesized that forested cover would be positively related to habitat quality and stability, and thus to genetic diversity, whereas developed and cropland cover would be negatively related. We estimated the magnitude of Pearson correlation coefficients (r) between diversity statistics and landscape characteristics and tested the null hypothesis that $r = 0$ based on two-tailed permutation tests (10^4 permutations per test) using custom scripts in R.

Mitochondrial analysis

Mitochondrial sequences were obtained for a subset of individuals that were randomly selected from each of the populations identified by the STRUCTURE analysis (see Results) and grouped by those populations for all subsequent analyses. We sought to obtain a minimum of five sequences from each population, and in cases where sample size fell below this value (e.g., Chestnut Creek) we sequenced all available individuals. Sequence data were analyzed in DNAsp version 5.10.1 (Librado and Rozas 2009) to estimate haplotype (Nei and Tajima 1981) and nucleotide (Nei 1987) diversity. Percent divergence was calculated between pairs of haplotypes by dividing the number of base-pair changes by the total number of sequenced base pairs. To infer evolutionary relationships between haplotypes, we used PopART (http://popart.otago.ac.nz) to create a median-joining haplotype network.

Results

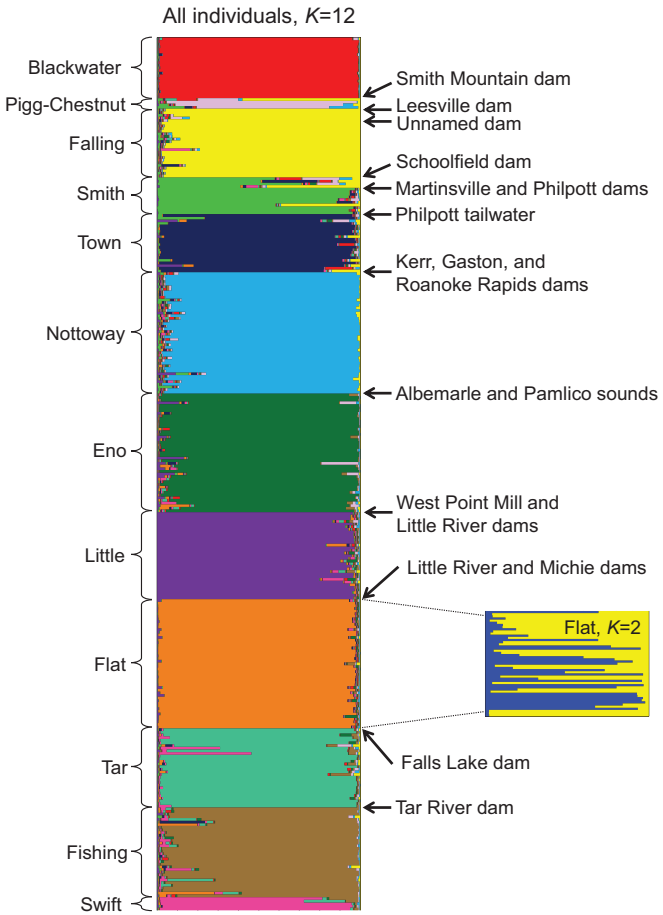
Utility of microsatellite markers

After Bonferroni correction, Hardy–Weinberg equilibrium was rejected in only one of 309 tests (Supplementary File 2[1]). Additionally, only 10 out of 3420 locus pairs showed evidence for linkage disequilibrium, and these pairs were randomly distributed among loci and sampling locations (Supplementary File 2[1]). These results indicated no strong influences of null alleles or linkage, so we retained all 19 loci for subsequent analyses.

Analysis of population structure

The STRUCTURE analysis of all 331 individuals indicated that a model of 12 discrete populations ($K = 12$) had the highest mean log-likelihood (Fig. 2). Simpler models with $K = 10$ also exhibited a

Fig. 3. Bar plot of individual assignments from the best-supported $K = 12$ STRUCTURE model based on analysis of the full dataset. Substructure analysis indicated that $K = 1$ within all of these clusters except Flat Creek (shown in inset), for which the best-supported $K = 2$. Post hoc population designations are indicated by brackets; locations of dams and other important barriers are indicated by arrows. [Colour online.]



high mean log-likelihood. However, the difference between $K = 10$ and $K = 12$ was biologically meaningful; the addition of these two clusters allowed for the separation of Pigg River – Chestnut Creek individuals from individuals captured in Smith River – Town Creek and the separation of Swift Creek from Fishing Creek individuals. The delta-*K* approach, on the other hand, indicated that $K = 2$ was the best-supported model (Fig. 2), a common outcome in other studies (Janes et al. 2017), which in our case left meaningful population structure undescribed. We therefore treated log-likelihood as the preferred method of evaluating model support and selected the $K = 12$ replicate with the highest log-likelihood as the best-supported model of *A. cavifrons* population structure. Based on this model, most populations corresponded to individual streams, and most boundaries between populations corresponded with presumed barriers to dispersal such as dams and basin boundaries (Fig. 3). The only exceptions to this pattern were (*i*) lower Smith River individuals clustered together with upper Smith River individuals, despite these areas being separated by Philpott Lake and two dams; (*ii*) lower Falling River individuals clustered with individuals from upper Falling River, despite these areas being separated by an unnamed dam near the Virginia State Route 40 bridge; and (*iii*) individuals from Fishing and Swift creeks were genetically distinguishable, despite the lack of a dam or basin boundary between these streams. The majority of individu-

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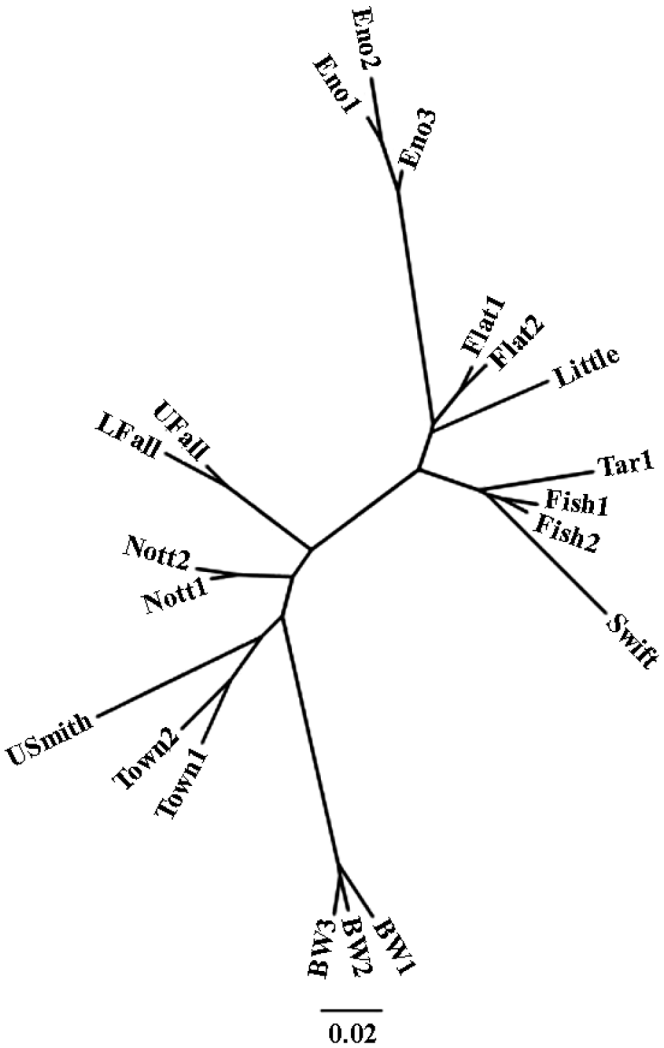
Fig. 4. Unrooted neighbor-joining tree based on microsatellite genetic distances (Nei's D_m) among the 20 sites with $n \geq 5$ individuals.



als shared most of their ancestry with other individuals from the same population, although there was evidence of admixture between sites in the Tar River and Fishing and Swift creeks (of which only Tar is separated by a dam).

In the hierarchical STRUCTURE analysis of substructure within each of these 12 clusters, the highest-log-likelihood K value was $K = 1$ in all populations except Flat Creek, for which the best-supported K_{subset} was $K = 2$ (Fig. 3). There was no apparent spatial or temporal component to the substructure detected within Flat Creek; clustering of individuals did not correspond to capture date or site (data not shown). Because the delta-K approach cannot evaluate support for a K value of $K = 1$ (Evanno et al. 2005) and showed little utility in the previous analysis, we did not use this metric to evaluate support for population substructure.

The site-based neighbor-joining tree supported population memberships indicated by the STRUCTURE analysis and provided additional information about the relative magnitudes of genetic differentiation among populations (Fig. 4). The tree was based on the 20 sites with $n \geq 5$ and possessed 11 terminal clusters of sites, which reflected the populations identified by the $K = 12$ model (sites in lower Smith River, lower Pigg River, and Chestnut Creek were excluded due to low sample sizes). The tree indicated a primary genetic division between basins draining to Albemarle Sound (Roanoke, Dan, Chowan) and Pamlico Sound (Tar, Neuse). Furthermore, streams in the Chowan, Dan, Neuse, and Tar basins tended to cluster with other streams in the same basin. In contrast, streams in the Roanoke basin did not cluster together; sites in Blackwater River were more genetically distant from upper and lower Falling River than from sites in the Dan and Chowan basins.

Table 2. Weighted parameter estimates from the two best-supported Bayesian generalized linear mixed models to explain variation in pairwise genetic differentiation (F_{ST}) among sites.

	Bayesian posterior estimates		
		95% credible interval	
Parameter	Mean	Lower	Upper
Intercept	0.006	−0.026	0.039
Distance	**0.036**	**0.008**	**0.063**
Barrier	**0.136**	**0.105**	**0.169**
Diversity	−0.129	−0.555	0.293
Distance × Barrier	−0.023	−0.049	0.005
Diversity × Barrier	**−0.847**	**−1.277**	**−0.466**

Note: Parameters with credible intervals that do not overlap zero are considered more strongly supported and are indicated in bold. Distance is in units of 100 km. Genetic diversity was scaled to have a mean of zero.

The AMOVA indicated that molecular variance components were significantly greater than zero (all $P < 0.0001$) at all scales considered, but spatial genetic variance was considerably greater among than within streams. The largest fraction of spatial molecular variation (11.0%) was among basins, whereas 8.3% of variation was among streams within basins, and only 1.0% was among sites within streams. The remaining 79.7% of variation was nonspatial, among individuals within sample sites. These findings were reinforced by pairwise estimates of F_{ST} among sample sites, which typically were small in comparisons within streams and (or) in the absence of a barrier. Pairwise F_{ST} values < 0.02 and not statistically different from zero ($P > 0.05$), thus indicative of panmixia, were found among eight site pairs located within the same stream (sites BW1 and BW2, BW2 and BW3, ENO1 and ENO2, ENO2 and ENO3, FISH1 and FISH2, FLAT1 and FLAT2, NOTT1 and NOTT2, and UFALL and LFALL). Of these comparisons, only UFALL and LFALL are separated by a dam. On the other hand, some site pairs within the same stream exhibited detectable genetic differentiation, despite the lack of a dam or other obvious barrier (sites BW1 and BW3, ENO1 and ENO3, and TOWN1 and TOWN2). For these site pairs, F_{ST} ranged from 0.028 to 0.040 (all $P < 0.05$).

The BGLMM indicated a significant role of distance-mediated dispersal (i.e., IBD), movement barriers, and genetic drift in promoting genetic differentiation among pairs of sites. Of the 13 models evaluated (Supplementary File 3[1]), two had model weights of at least 0.01 and were included in our confidence set (Table 2). The best-supported model had a DIC weight of 0.64 and featured parameters describing effects of spatial distance, barriers, genetic diversity, a distance × barrier interaction, and a diversity × barrier interaction. The second-best-supported model had a weight of 0.36 and included all of these effects except the distance × barrier interaction. Based on parameter estimates weighted across these two models, a composite model exhibited good fit to the observed data, with a pseudo-R^2 of 0.74 (Fig. 5). Most riverine-connected sites exhibited small observed and predicted F_{ST} values (<0.04), whereas most sites separated by a barrier exhibited observed and predicted F_{ST} values > 0.10 and up to 0.42. The only substantial outlier was the comparison of upper to lower Falling River (above versus below an unnamed dam), which exhibited a predicted F_{ST} of 0.12 but an observed $F_{ST} < 0.001$.

Some effects included in the composite model exhibited a stronger influence on F_{ST} than others. The model intercept was approximately zero, indicating that in the absence of distance or

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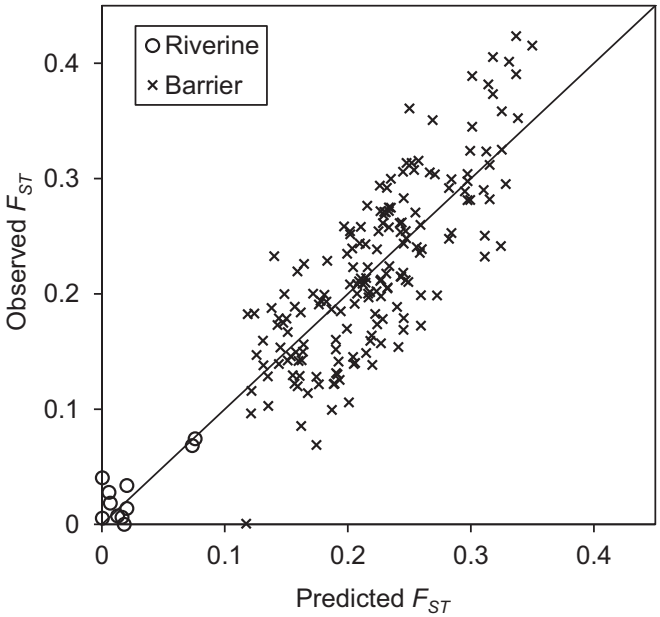
Fig. 5. Relationship between observed and model-predicted F_{ST} values in the best-supported Bayesian generalized linear mixed model, plotted separately for sites separated by barriers versus completely riverine connections. A 1:1 line is shown for comparison. Based on concordance of predicted and observed values, the model's pseudo-R^2 was 0.74.



barriers, two sites with average heterozygosity would exhibit an $F_{ST} \sim 0$. The distance effect indicated that in the absence of a barrier, F_{ST} would increase by 0.036 for every 100 km of riverine distance. Although the presence of a distance × barrier effect indicated that IBD was weaker for sites separated by a barrier, the confidence interval (CI) for this term overlapped zero, indicating weak support. In contrast, the barrier effect strongly influenced differentiation; F_{ST} was predicted to increase by 0.136 when a barrier was present between sites, equivalent to a riverine distance of 378 km. Barriers also mediated the effect of genetic diversity. In the absence of a barrier, genetic diversity was negatively but weakly related to F_{ST}, with a CI that overlapped zero. However, in site pairs separated by a barrier, the negative relationship between diversity and F_{ST} was well supported, with a CI that did not overlap zero. The model predicted that for such site pairs, a decrease in mean H_E of 0.1 would result in an increase in F_{ST} of approximately 0.1.

Contemporary dispersal among populations

Testing for first-generation migrants in STRUCTURE revealed no individuals assigned with high probability to a different population than the one from which they were collected, despite the assumption of a liberal migration prior of 0.5. Thus, not surprisingly, the same result was obtained with more conservative priors (results not shown). Most of these populations are separated by some type of movement barrier such as a dam or marine habitat. However, even populations not separated by an obvious physical barrier (e.g., Fishing and Swift creeks) had no detectable dispersal events.

Genetic diversity of populations

Genetic diversity statistics were calculated for each population delineated by the STRUCTURE clustering analysis, except that we further separated upper and lower Smith rivers for this analysis, based on other evidence that these areas are genetically distinct (see Discussion). The A_4 and H_E statistics were strongly positively correlated ($r = 0.715$, $P = 0.003$), such that population rankings for both of these diversity measures were similar. Both were lowest in the Blackwater population and highest in the Pigg population (Table 3). The M ratio ranged from 0.657 in Flat to 0.843 in Nottoway (Table 3) and was not strongly correlated with either A_4 ($r = 0.467$, $P = 0.096$) or H_E ($r = 0.276$, $P = 0.227$). Of the nine populations for which M could be calculated based on sample size requirements, only the Nottoway population exhibited an M within the published range of values (0.823 to 0.926) from populations with known demographically stable histories (Garza and Williamson 2001). In contrast, populations in Flat Creek and Eno River exhibited M values within the published range (0.599 to 0.693) from populations known to have experienced a bottleneck (Garza and Williamson 2001), and the other six populations' M values were intermediate to these two ranges.

Estimates of N_e depended on the rare-allele-frequency cutoff used, but the order of magnitude and ranking of populations was similar across these modeling choices (Table 3). In a number of cases, upper 95% confidence limits of N_e were infinite, and in two cases (Blackwater for 0.02 cutoff and Tar for 0.05 cutoff), estimates of mean N_e were infinite. Infinite estimates can occur due to inadequate sample size or large true N_e, as N_e becomes difficult to estimate under either condition (Waples and Do 2010). In our case, because infinite estimates did not appear to relate to sample size, we interpret them as indicative of large true N_e. Regardless, all populations exhibited an estimable mean for at least one of the two cutoff values. Estimates of mean N_e were greatest for the Tar ($N_{e0.02} = 3709$; $N_{e0.05}$ = infinite), Nottoway ($N_{e0.02} = 437$; $N_{e0.05} = 196$), and Blackwater populations ($N_{e0.02}$ = infinite; $N_{e0.05} = 168$), whereas the other six populations exhibited one or more mean N_e estimates of <100 individuals, and populations in Little River and Flat Creek exhibited mean $N_e < 50$ for both cutoffs. Moreover, lower 95% confidence limits were <50 for all populations except Tar, Nottoway, and Fishing.

Watershed area ranged among populations from 84 to 2927 km^2 (Supplementary File 4[1]) and was significantly positively correlated with all three genetic diversity measures (A_4, H_E, and M; Table 4). Percent coverages of alternative land uses varied substantially among populations. Forested, cropland, and developed land cover in 2011 ranged among populations from 56% to 82%, 9% to 29%, and 3% to 14%, respectively (Supplementary File 4[1]). On the other hand, land-use change between 2001 and 2011 was modest. All populations exhibited decreases (1% or less) in cropland cover, all populations except Fishing exhibited decreases (1%–4%) in forested land cover, and all populations except Little exhibited increases (1% or less) in developed cover. In general, land use correlated more strongly with M than with the other two diversity measures. A_4, H_E, and M all were positively correlated with percent forested area and negatively correlated with percent cropland and percent developed area (Table 4), but of these, the only statistically significant correlation was between M and percent cropland ($r = -0.607$, $P = 0.045$). All diversity statistics also were negatively correlated with Δpercent cropland and Δpercent developed over time (Table 4), but of these, the only statistically significant correlation was between M and Δpercent developed ($r = -0.574$, $P = 0.045$). Correlations with Δpercent forested were inconsistent and nonsignificant.

Assessment of mitochondrial diversity

Among the 64 sequenced *A. cavifrons* individuals, 12 unique haplotypes were identified (Table 5). Although nuclear analyses indicated that upper and lower segments of Falling River (above and below an unnamed dam) were part of the same population, they exhibited substantial mtDNA differences, so we treated them separately for mtDNA analyses. The lower Falling population exhibited the greatest number of haplotypes (4), haplotype diversity (0.9), and nucleotide diversity (0.002). In contrast, upper Smith, Eno, and Swift exhibited no mtDNA variation across the five individuals analyzed in each of these populations. Private haplotypes

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Table 3. Nuclear genetic diversity statistics for each inferred *A. cavifrons* population, across 19 microsatellite loci.

Population	n	A_4	H_E	M	N_e (95% CI), 0.02 cutoff	N_e (95% CI), 0.05 cutoff
Blackwater	23	2.03 (0.75)	0.50 (0.15)	0.749 (0.221)	inf (32.9, inf)	168.3 (19.4, inf)
Pigg	4	3.74 (1.89)	0.76 (0.19)	—	—	—
Falling	26	2.88 (1.33)	0.60 (0.26)	0.708 (0.232)	133.3 (50.9, inf)	70.2 (31.3, inf)
Upper Smith	10	2.89 (1.47)	0.65 (0.22)	—	—	—
Lower Smith	4	2.71 (1.26)	0.69 (0.23)	—	—	—
Town	22	2.89 (1.32)	0.61 (0.22)	0.750 (0.199)	79.8 (34.6, inf)	40.2 (18.8, 374.8)
Nottoway	46	3.40 (1.41)	0.66 (0.21)	0.843 (0.169)	436.9 (159.1, inf)	196.4 (89.1, inf)
Eno	45	2.66 (1.01)	0.51 (0.24)	0.693 (0.206)	77.4 (37.8, 421.3)	67.6 (31.7, 428.7)
Little	33	2.88 (1.35)	0.55 (0.27)	0.738 (0.161)	43.2 (27.5, 81.4)	31.4 (19.4, 60.3)
Flat	49	2.84 (1.16)	0.63 (0.18)	0.657 (0.181)	41.9 (22.5, 107.9)	35.6 (19.7, 81.9)
Tar	30	2.95 (1.46)	0.57 (0.28)	0.773 (0.205)	3709.0 (118.9, inf)	inf (138.2, inf)
Fishing	34	3.10 (1.60)	0.59 (0.30)	0.811 (0.811)	161.2 (75.5, inf)	99.5 (52, 458.7)
Swift	5	2.53 (1.20)	0.61 (0.20)	—	—	—

Note: Allele richness per locus standardized to a sample size of four individuals (A_R), expected heterozygosity (H_E), and the ratio of allele richness to allele size range (M) were averaged across loci (standard deviations in parentheses). Means and 95% confidence intervals (CIs) of effective population size (N_e) were estimated for populations with $n \geq 20$ samples, based on exclusion of rare alleles that occurred at frequencies of less than 0.02 or 0.05. Estimates of N_e were sometimes infinite (inf). The Pigg population included individuals from Chestnut Creek. Dashes indicate statistics not estimated due to sample size.

Table 4. Pearson's correlation (r) among allele richness per locus standardized to a sample size of four individuals (A_4), expected heterozygosity (H_E), and the ratio of allele richness to allele size range (M) and indices of habitat size and quality, including watershed area, watershed percent coverage of various land-cover types in 2011, and the percent change (Δ%) in these coverages between 2001 and 2011.

	A_4	H_E	M
Watershed area	**0.657**	**0.588**	**0.730**
% Forested	0.095	0.173	0.463
% Cropland	−0.473	−0.324	**−0.607**
% Developed	−0.094	−0.392	−0.463
Δ% Forested	0.031	−0.281	0.312
Δ% Cropland	−0.544	−0.447	−0.557
Δ% Developed	−0.223	−0.322	**−0.574**

Note: Correlations where r is significantly different from zero are indicated in bold.

were observed in Fishing Creek, Nottoway River, Blackwater River, lower Falling River, Chestnut Creek (Pigg population), and lower Smith River (Table 6). However, based on the median joining haplotype network, all haplotypes had close evolutionary relationships with each other, being separated by only one to five mutation events (0.1% to 0.5% divergence; Fig. 6). There was no clear geographic structure to haplotype relationships, and no basin clustered entirely separately from the rest. Three of the five basins (Neuse, Dan, and Roanoke) shared at least one haplotype with all other basins, and the Chowan and Tar basins shared at least one haplotype with all basins but each other (Fig. 6).

Discussion

Population structure, dispersal, and gene flow

An understanding of how populations are structured is crucial for the development of plans to monitor, manage connectivity, and ensure the evolutionary potential of populations and ensuring that these activities are conducted at the appropriate spatial scales. We found that the primary determinant of contemporary population structure for *A. cavifrons* was the distribution of anthropogenic barriers to fish movement. All but one of the population boundaries we detected corresponded with at least one dam or tailwater. We found no evidence for contemporary dispersal or gene flow among populations separated by these barriers. This corroborates a number of previous studies of riverine fishes that have demonstrated a profound effect of dams and associated reservoirs on genetic connectivity (Pritchard et al. 2007; Raeymaekers et al. 2008; Skalski et al. 2008). Although the effect of tailwaters on fish dispersal has been less investigated, the rarity of this species in the tailwaters below Leesville and Philpott dams (Scott Smith, VDGIF, personal communication) suggests that it avoids these hydrologically unstable habitats. Moreover, the strong differentiation (F_{ST} = 0.147) of Town Creek from nearby lower Smith River suggests that the hypolimnetic release from upstream Philpott dam acts as a thermal barrier to dispersal, similar to Roberts et al.'s (2013) findings for Roanoke logperch (*Percina rex*). Range fragmentation has two primary management implications for *A. cavifrons*. First, there is no potential for natural rescue or recolonization among fragmented areas. Second, fragmentation, combined with small population size and rapid genetic drift (see below) has dramatically inflated genetic divergence among populations; we estimated that the presence of a dam increased F_{ST} by the same amount as 378 km of riverine distance. The evolutionary consequences of this anthropogenically accelerated divergence are unknown, but bear concern when formulating policies to manage connectivity for the species.

Whereas anthropogenic barriers had profound effects on population structure, in the absence of such barriers we generally saw weak genetic differentiation and little structure. Reaches within the same stream were genetically similar, suggesting that natural habitat boundaries (e.g., channel units and stream junctions) have had little effect on *A. cavifrons*' dispersal. Geographically extensive panmixia (up to 64 river km in Nottoway) contrasts with conventional wisdom that stream fishes in general, and centrarchids in particular, seldom move beyond reach boundaries (Gerking 1953). Although no previous studies have assessed the movement patterns of *A. cavifrons*, Gatz and Adams (1994) found the close congener *A. rupestris* to be highly sedentary, with a maximum detected movement of only 600 m and only about one-third of recaptured fishes moving more than 100 m over a 3-year period. Mark–recapture study biases may result in underestimates of the distance and frequency of fish movement (Albanese et al. 2003), whereas genetic studies of the same taxa can reveal watershed-scale dispersal and effective gene flow. For example, mark–recapture studies of *P. rex* indicated frequent short-distance (<100 m) movements and infrequent long-distance movements (>1 km; Roberts et al. 2008), whereas genetic analyses of this same population revealed fre-

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Table 5. Estimates of mitochondrial genetic diversity for populations of *A. cavifrons*, based on the cytochrome *b* gene.

Population	*n*	Haplotypes	Private haplotypes	Segregating sites	Haplotype diversity	Nucleotide diversity
Blackwater	5	2	1	1	0.400 (0.237)	0.00041 (0.00024)
Lower Falling	5	4	2	4	0.900 (0.161)	0.00206 (0.00053)
Upper Falling	5	2	0	1	0.600 (0.175)	0.00062 (0.00018)
Upper Smith	5	1	0	0	0	0
Town	5	3	0	2	0.800 (0.164)	0.00103 (0.00028)
Nottoway	5	3	2	3	0.700 (0.218)	0.00165 (0.00051)
Eno	5	1	0	0	0	0
Flat	5	2	0	1	0.400 (0.237)	0.00041 (0.00024)
Little	5	2	0	1	0.400 (0.237)	0.00041 (0.00024)
Tar	5	2	0	1	0.400 (0.237)	0.00041 (0.00024)
Fishing	5	2	1	2	0.600 (0.175)	0.00123 (0.00036)
Swift	5	1	0	0	0	0

Note: Standard deviations are displayed in parentheses. Upper and lower Falling River were determined to be part of the same population but were treated separately for this analysis.

Table 6. Frequency of the 12 deduced mitochondrial cytochrome *b* haplotypes (lettered A through Q), by stream system.

	Basin													
	Neuse			Tar			Chowan	Roanoke			Dan			
Haplotype	ENO	FLT	LIT	TAR	FIS	SFT	NOT	BLW	UFL	LFL	CHT	USM	LSM	TWN
A	5	4	4	4				1	2	2		5	1	2
B		1	1											
C					2									
D								4						
E				1	3	5								2
F							3							
J										1				
K										1				
L											2			
M							1							
P							1		3	1				1
Q													1	
Total	5	5	5	5	5	5	5	5	5	5	2	5	2	5

Note: Abbreviations for sample sites are as follows: ENO = Eno River, FLT = Flat River, LIT = Little River, TAR = Tar River, FIS = Fishing Creek, SFT = Swift River, NOT = Nottoway River, BLW = Blackwater River, UFL = upper Falling River, LFL = lower Falling River, CHT = Chestnut Creek, USM = upper Smith River, LSM = lower Smith River, and TWN = Town Creek.

quent dispersal and gene flow throughout a 60 km reach (Roberts et al. 2016). These considerations aside, the rate at which *A. rupestris* has invaded new waterbodies in Virginia (Eschenroeder and Roberts 2018) provides strong indirect evidence that *Ambloplites* species are capable of rapid, extensive dispersal. From a management standpoint, high connectivity in the absence of a barrier implies that whole streams and watersheds, not individual reaches, should be the focal points of monitoring and management actions (Fausch et al. 2002).

Although barrier presence–absence had the largest impact on effective connectivity for *A. cavifrons*, it is important to note that the juxtaposition of barriers did not perfectly align with the juxtaposition of population boundaries. The few exceptions merit closer examination, because they lend additional insight into the habitat ecology of the species. First, despite the lack of an intervening dam, we detected a population boundary between Fishing and Swift creeks. Occupied sections of these streams are separated by naturally low-gradient, swampy sections in their lower reaches, and these unsuitable habitats may discourage transit and preclude the exchange of migrants. Second, we also detected modest population substructuring within some streams that lacked dams, as evidenced by $F_{ST} > 0$ or $K_{subset} > 1$. We attribute modest but detectable IBD in Blackwater, Eno, and Town to fine-sediment deposition in low-gradient sections of these streams (Jenkins and Burkhead 1994), which attenuates dispersal between more suitable, occupied reaches. The apparent substructure within Flat Creek, on the other hand, was nonspatial (i.e., not among sites) and was detected only in the STRUCTURE analysis. Although we cannot confidently interpret this result, we assume that apparent substructure was due to an Allendorf–Phelps effect, wherein two differentiated families or cohorts were disproportionately represented in our samples (Allendorf and Phelps 1981). The chance of statistically detecting such temporal and (or) family structure would be elevated in a small population like Flat, where mating with close relatives is more likely. Third, two putative barriers to movement did not consistently result in detectable population boundaries. Despite being separated by Philpott Reservoir and Dam, STRUCTURE indicated that upper and lower Smith rivers were part of the same population. However, this STRUCTURE result may have been an artifact of small sample size in lower Smith ($n = 4$), given that pairwise differentiation between these stream segments was large ($F_{ST} = 0.141$) and significant ($P < 0.0001$). Another putative movement barrier was the 4 m high mill dam separating upper and lower Falling rivers. Although none of the nuclear analyses (STRUCTURE, neighbor-joining tree, F_{ST}) provided any evidence that this dam created population structure, mtDNA haplotype diversity was substantially higher downstream than upstream of the dam. This may be due to a source–sink dynamic occurring between the two stream segments. It has been demonstrated by previous studies that obstructions in the river may allow for unidirectional flow of individuals and alleles from upstream sites (Meeuwig et al. 2010; Meldgaard et al. 2003; Neraas

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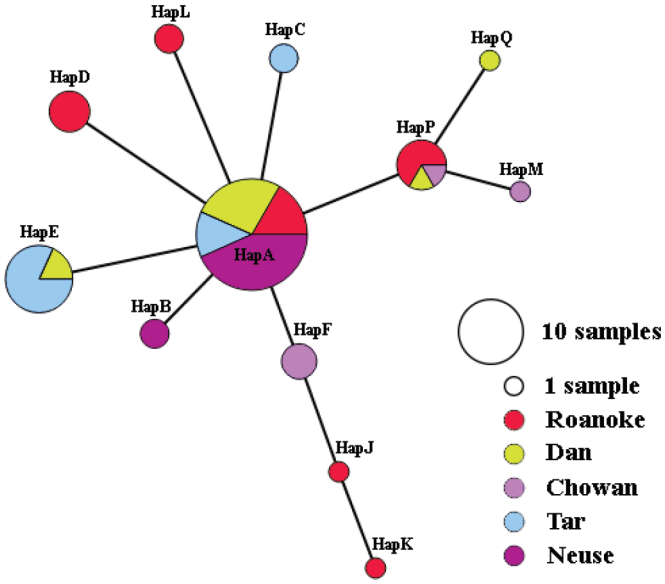
Fig. 6. Median-joining haplotype network depicting relationships among the 12 deduced *A. cavifrons* mitochondrial haplotypes (indicated by lettered circles). Each line segment indicates a single hypothesized mutation event. Circle size is proportional to the number of individuals possessing that haplotype, and circle colour indicates basin of origin. [Colour online.]



and Spruell 2001). In further support of the one-way migration idea, the river below the dam contained a mix of *A. rupestris*, *A. cavifrons*, and hybrids, whereas no *A. rupestris* or hybrids have been detected above the dam (Eschenroeder and Roberts 2018).

Results of this study indicate that the contemporary population structure of *A. cavifrons* is highly fragmented, yet we also found evidence that historical gene flow was more extensive, both within and between river basins. The BGLMM results indicated weak IBD, once the effects of drift on heterozygosity were accounted for, regardless of dams or watershed divides. A steep IBD slope would indicate either that historical gene flow was geographically limited or that contemporary fragmentation and drift have made it appear so (Hutchison and Templeton 1999). In contrast, the observed shallow IBD slope presumably reflects geographically extensive historical gene flow that is still detectable in the nuclear data despite recent fragmentation. Based on weighted estimates from best-fitting BGLMM models, we estimated that nearly 300 km of riverine distance would be required to attain an F_{ST} of 0.1. Mitochondrial DNA also bears the mark of high historical connectivity for *A. cavifrons*. There was a high degree of haplotype sharing among basins (i.e., lack of fixation), and the overall nucleotide divergence among haplotypes was low (<0.5%).

Factors influencing the genetic diversity and viability of populations

Depending on the criteria used, we delineated 12 to 13 extant populations of *A. cavifrons*, all of which appear completely isolated from conspecifics. Indices of genetic diversity varied widely among these population fragments, at least in part due to variation in contemporary patch size and habitat quality. Consistent with theoretical expectations and previous study findings (Koizumi et al. 2006; Raeymaekers et al. 2008), watershed area was positively correlated with all three diversity indices (H_E, A_4, M). This suggests that larger watersheds housed larger, more stable populations, because they featured a greater extent of accessible habitat, greater habitat volume, a greater "portfolio" buffering effect against disturbance, or a combination of these features (Taylor and Warren 2001; Fagan 2002; Morita and Yamamoto 2002). In 17 of 18 comparisons, diversity indices also tended to correlate with watershed land use in the directions we hypothesized (positively for forest, negatively for urban and agriculture), though only two of these tests were statistically significant. Only nine populations had sample sizes large enough to run these tests, which limited our statistical power to detect an effect. Nonetheless, the directions of relationships were consistent with our general hypothesis that human alteration of watershed land cover would reduce habitat quality instream and thereby reduce population size and (or) stability for *A. cavifrons*. As previously described, the species appears intolerant of hydrologic instability and excess fine-sediment deposition, two common outcomes of urban and agricultural development (Allan 1995; Walsh et al. 2005). Cropland area tended to be most strongly associated with diminished genetic diversity. This does not necessarily mean that agricultural practices have had a larger effect on *A. cavifrons* populations, but rather may reflect that agricultural land use is much more common than urban land use in all assessed watersheds. Additionally, all land-use types exhibited some correlation with one another, and thus relationships of genetic variables to all three were to be expected. Patterns in correlation between genetic variables and percent change in watershed land use were less pronounced. This may be because land-use change was measured over only a 10-year period (2001 to 2011), likely representing only a fraction of the changes that have taken place over the past century and covering few generations (*A. cavifrons* generation time is likely ~3 years; Jenkins and Burkhead 1994). In fact, the land-use change in most watersheds was slight, averaging about 1.5% loss of forest cover. If we were to assess the relationship between population size and diversity and land-use changes over a broader time span that encompassed the substantial growth of the human population in the region (i.e., a span beginning prior to 1950), correlations may have been stronger. Although we cannot parse the potential effects of angling harvest on genetic diversity from the effects of land use, the two sampled populations that experience the greatest fishing pressure (Nottoway and Eno) showed no consistent signal of reduced diversity or bottlenecks, suggesting that fishing has had minimal effects.

We also noted that of the genetic statistics examined, M always was the index most strongly correlated with a given landscape variable. Whereas the other two variables serve as indicators of historical effective population size and change relatively slowly following population-size changes (Schwartz et al. 2007), M is a proxy for rapid population-size change in the recent past (Garza and Williamson 2001). Thus, our findings suggest that relative to H_E and A, M might be a more timely, sensitive indicator of recent, anthropogenically induced changes to population size. It is worth noting that the correlations between genetic diversity and land use may be a legacy effect from intense deforestation that occurred over the past two centuries rather than the result of more recent land-use change, but in any case the data do not suggest that contemporary improvements in land management practices have resulted in any concomitant improvements in the genetic diversity of *A. cavifrons*.

Given the combined effects of habitat fragmentation, reduced patch size, and reduced patch habitat quality, we would expect populations of *A. cavifrons* to show reductions in effective population size, potentially in direct proportion to the magnitude of these effects. Our results support the general notion that many extant populations have small effective sizes that do not pass common conservation criteria for long-term evolutionary viability. Franklin (1980) proposed the general guidelines that populations should maintain $N_e > 50$ over the short term to avoid the risk of inbreeding depression and $N_e > 500$ to maintain genetic diversity in the long term. As we lack species-specific criteria, these rules of thumb provide benchmarks against which to compare our estimates of N_e for *A. cavifrons*. Based on the most conservative estimate of mean N_e for each population (i.e., the allele-frequency

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cutoff resulting in the smallest mean N_e), three populations (Town, Little, and Flat) did not pass the short-term threshold of $N_e > 50$, and five of the remaining six evaluated populations did not pass the long-term threshold of $N_e > 500$. Only Tar River, which had mean N_e estimates of 3709 (0.02 cutoff) and infinity (0.05 cutoff) was secure according to the "50:500 rule".

The model implemented by LDNe to estimate N_e assumed random mating, nonoverlapping generations, and sampling of a single generation (Waples and Do 2008). We violated the latter two assumptions, because *A. cavifrons* is iteroparous and we pooled multiple (unknown-age) cohorts for analysis. However, all populations should have been affected similarly by any biases this created (e.g., Luikart et al. 2010), and thus the variation in N_e estimates between populations should be proportional to the relative variation in the actual size of the gene pools (Robinson and Moyer 2013). Furthermore, simulations suggest that violating this assumption creates only a slight upward bias (i.e., value inflated by less than 10) in estimates of N_e (Waples 2006). Another issue was the fact that relatively small sample sizes for some populations reduced the precision of our estimates and resulted in frequent infinite upper bounds and occasional infinite means. As such, it may be prudent to construe our estimates as relative measures of approximate gene-pool size that are intermediate to N_e and the effective number of breeders (N_b), rather than exact measures of N_e. Taking this approach, these estimates provide a method of ranking populations from smallest to largest effective size, which may aid in the identification of populations that may benefit from future translocation efforts and which populations may serve as the best sources for translocated individuals. Even in relative terms, N_e varies by nearly two orders of magnitude between Tar River and the smallest populations considered. Thus, larger populations like Tar and Nottoway may prove important reference points for and potential donors to less secure populations like Little River, Town Creek, and Flat Creek.

Comparison of N_e estimates to the bottleneck index M provided additional insight that neither statistic provided alone. Eno River and its tributaries Little River and Flat Creek all harbored populations with relatively small N_e and low M. This suggests that low N_e throughout this system was due to a relatively severe and sudden bottleneck within the past 6–50 generations (Garza and Williamson 2001; Peery et al. 2012), a time period that coincides with dam construction and increasing urbanization in this region. In fact, the Eno watershed contains nearly twice the developed land area of any other watershed examined (14.1%; Supplementary File 4[1]). In contrast, the population in Town Creek had a relatively small N_e but not a particularly low M, suggesting that N_e has been small but stable over the recent past there.

Potential recovery actions

Ambloplites cavifrons currently exists in a mosaic of fragmented populations, and our analysis suggests that at least some of these populations may be incapable of maintaining long-term genetic variability. Additionally, a lack of connectivity means populations that decline cannot be naturally rescued by migrants from other populations, and thus the fate of each will be decided by local habitat conditions (Danancher et al. 2008). The suite of changes associated with the construction of dams and the conversion of forested watersheds into cropland and urban centers, including alterations to the natural flow regime, increased sedimentation, and pollution (Poff et al. 1997; Walsh et al. 2005), acts to alter the ecosystem to which stream biota are adapted. However, anthropogenic effects are not limited to chronic impacts and can include stochastic events like pollution-induced fish kills. Indeed, streams inhabited by *A. cavifrons* have experienced major fish kills from chemical discharge. Sudden population declines from stochastic events may result in population bottlenecks and associated demographic instability and further genetic diversity loss. There are no reliable historical data on chemical spills and fish kill events, and as such it is not possible to parse the effects such events may have had on contemporary genetic diversity from the chronic effects of land use change. However, it is likely that the incidence of such events has increased and will continue to increase with expanding urban development.

Connectivity could be improved through the removal of barriers, but this tactic increases the risk of spread of invasive *A. rupestris* (Eschenroeder and Roberts 2018). Alternatively, the translocation of individuals could be carried out in a pattern that mimics historical connectivity (DeHaan et al. 2011). This could improve the genetic diversity of isolated populations and aid in the recovery of populations that have suffered declines. Translocation carries some risk of outbreeding depression, which may result from the disruption of coadapted gene complexes and local adaptations (Edmands 2007; George et al. 2009). This risk seems low in *A. cavifrons* given that the mixing of fish from the Tar drainage of North Carolina with fish native to Town Creek in the Dan drainage of Virginia did not result in any detectable declines. However, nuclear genotypes provide little evidence that the fish introduced to Town Creek integrated, and the stocking the Pigg River did not result in any detectable increases in abundance in that system (Jenkins and Cashner 1983). This suggests more thorough post-stocking monitoring studies are necessary to evaluate the ability of introduced individuals to successfully integrate. Translocation may be particularly important for populations that have experienced declines and (or) have relatively low N_e. Specifically, populations in the Neuse basin may benefit from reciprocal translocation efforts, as they were found to have the smallest effective sizes and there is evidence that they have experienced bottlenecks. Town Creek had the next smallest effective size, and the translocation of individuals from within the Dan basin (upper and lower Smith River) may also be warranted, but the collection of more samples from the Smith to allow for estimates of N_e and the detection of recent declines is advisable. Although N_e of the Tar River and Fishing Creek populations are high relative to other populations, more samples are needed from Swift Creek to determine its status. High N_e and genetic diversity along with a lack of barriers preclude the need for translocations in the Chowan basin. In addition to the supplementation of existing populations, managers might consider opportunities to introduce or reintroduce the species into suitable habitats within its presumed natural range but where it does not currently occur. Such an approach may help prevent extirpation from the Virginia portion of its range, which has been greatly diminished by invasive *A. rupestris* (Eschenroeder and Roberts 2018). Although these potential restoration efforts are being considered by state managers, no specific management actions are currently planned.

For efforts to boost genetic diversity and for N_e to be effective, the factors causing loss of genetic diversity must be addressed (Caro and Laurenson 1994). The correlation of decreases in genetic diversity to deforestation and increases in cropland and urban development suggest that land-use practices have contributed to a gradual decline of *A. cavifrons* populations. Thorough assessments of the contemporary quantity and quality of *A. cavifrons* habitat are warranted, as knowledge of these parameters would help to guide restoration efforts to improve habitat, to indicate whether streams with diminished populations can support supplementation with translocated individuals, and to determine whether currently uninhabited streams could support new populations. We posit that only through a combination of tactics, including restoring habitat, limiting the spread of invasive competitors, and reestablishing historical connectivity, can the future of *A. cavifrons* be assured.

Acknowledgements

We thank M. Pinder and C. Oakley for bringing this project to fruition and for invaluable assistance with field work and logistics. R. Adams, B. Blood, E. Brittle, P. Chrisman, N. Kmetz, B. Parks,

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B. Plunkett, S. Smith, G. Strickland, and D. Wheaton helped collect the samples. B. Plunkett, S. Harrison, and D. Kessler assisted with laboratory work. P. Angermeier, E. Hallerman, S. Harrison, and S. Vives provided helpful feedback on early versions of the manuscript. This work was funded by VDGIF and NCWRC. Fish collections were made under the auspices of Virginia Tech Institutional Animal Care and Use Committee protocol 12-193-FWC. The authors declare there is no conflict of interest in this article.

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